Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework in this chapter. A conforming mixed finite element discretization (MFEM) is proposed to solve the static Darcy-Stokes coupled methcnics system. The normal flux are captured at the edges of element by the MFEM scheme. The coupled weak form is also adjusted to be locally mass conservative. We solve the transport system with finite volume method (FVM), which is naturally a locally mass conservative scheme.

In this chapter, we first introduction the mesh used in our computation. We present two stable pair for Darcy and Stokes individually equipped with the direct serendipity space. We then apply the defined discrete space to discretize the coupled system. After that, we discuss a noval nonlinear WENO weighting scheme for FVM. Error estimate will be given to MFEM and FVM scheme.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred to as an element in finite element context, while called as a cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point.

We assume shape regularity for our quadrilateral mesh. A formal definition of uniform shape regularity of a quadrilateral mesh is given according to Girault and Raviart (2011).

Definition 1.1. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles $T_i, \quad i \in \{0, 1, 2, 3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilaterl E, and use the remaining three vertex. We define the following parameters as

$$
\begin{aligned}
h_E &= \text{diameter of E}, \\
\rho_E &= 2 \min_{1\le i\le 4}\{\text{diameter of largest circle inscribed in } T_i\}.
\end{aligned} \tag{1.1}
$$

A mesh partion $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$ we have the the ratio lower bound as

$$
\rho_E/h_E \ge \sigma_* > 0, \tag{1.2}
$$

where the shape regularity parameter σ^* is independent of $\mathcal{T}_h$.

In Fiugre 1.1, we show a reference element (cell) $\hat{E} = [-1, 1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element (cell) E. The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0, 1, 2, 3\}$ both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also start from zero. The vertices can be understood as $v_i = e_i \cap e_{i+1}$, and $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, e.g., the length of edge e_1 in Figure 1.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Not that the unit normal and tangent vectors are invariant under scaling mapping H.

For the ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. In Figure 1.2, we illustrate a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{T}_h$, which is

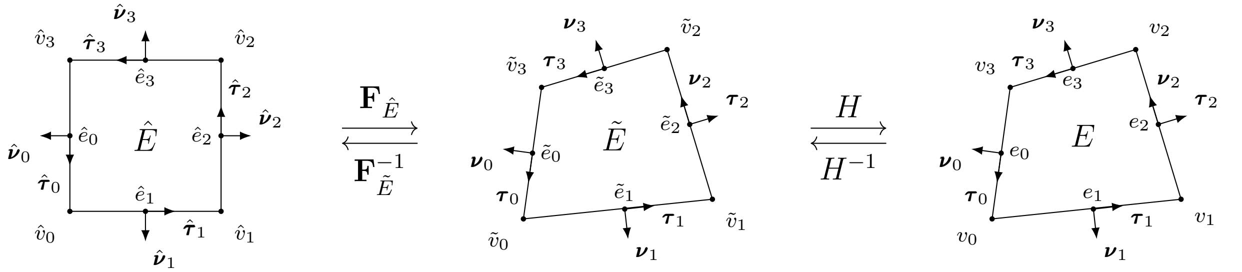


Figure 1.1: An exhibition of a reference square and general non-degenerate quadrilateral element(cell) before and after scaling.

generated by randomly distorting the interior vertex of a square mesh. Note that vertex on the boundary $\partial\Omega$ remain untouched.

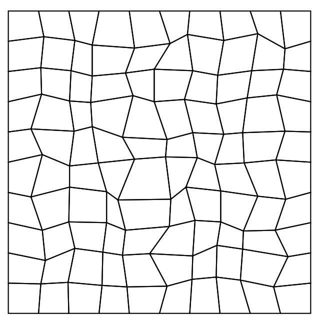


Figure 1.2: An exhibition of a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{T}_h$.

1.2 Mathematical Preminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practical, our discussion are confined to two dimension cases. We work only with real-valued functions for simplicity.

To begin with, we recall the definition of a Sobolev space.

Definition 1.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \leq p \leq \infty$, and $m \geq 0$ be an integer.

The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq m\}, \tag{1.3}$$

where the multi-indices $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-indices derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \{\int_\Omega |f|^p\}^{1/p}. \tag{1.4}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 1.3. For $f \in W_p^m(\Omega)$, Sobolev norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \{\sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^p(\Omega)}^p\}^{1/p}, \quad p < \infty, \tag{1.5}$$

and

$$\|f\|_{W_\infty^m(\Omega)} = \max_{|\alpha| \leq m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p = \infty. \tag{1.6}$$

The Sobolev seminorm $|\cdot|_{W_p^m}$ is

$$|f|_{W_p^m(\Omega)} = \{\sum_{|\alpha| = m} \|D^\alpha f\|_{L^p(\Omega)}^p\}^{1/p}, \quad p < \infty. \tag{1.7}$$

In the special case when $p = 2$, we denote $W_2^m = H^m$. The corresponding norm and seminorn are written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$ space

Theorem 1.1. *(Trace Theorem)*

Let k and d be integers with $0 < k < d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s > k/2$.

Proof. See Arbogast and Bona (2025). $\square$

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{1.8}$$

We use the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 1.2. *(Bramble-Hilbert Argument)*

Let $\Omega \in \mathbb{R}^d$ be a domain, and for function $f \in W_p^m(\Omega)$, $m \geq 0$, $1 \leq p < \infty$. There exists a constant C such that the inequality holds, i.e, that

$$\inf_{p \in \mathbb{P}_{k-1}(\Omega)} \|f - p\|_{W_p^m(\Omega)} \leq C h^{k-m} |f|_{W_p^k(\Omega)}, \quad j = 0, 1, \ldots, k, \tag{1.9}$$

where the constant C has an upper bound due to shape regularity assumption.

Proof. See Dupont and Scott (1980b). $\square$

1.3 Direct Serendipity Space

Serendipity element have the same accuracy of approximation with classical tensor product element while use smaller number of degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\mathrm{div})$ conforming direct mixed finite elemenent space are constructed with the direct serendipity function space.

To begin with, we introduce some distance auxilliary functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.10}$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity. The value of distance function λ_i keeps positive as long as $\boldsymbol{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{1.11}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, $\mathbf{x}_{d,i}^* = \mathbf{x}_{v,i}$, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i.

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.3.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E), r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}, \quad r \geq 2$. We present the second order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.12}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}, \tag{1.13}$$

where $i \equiv i \pmod 4$. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.14}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.15}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0, 1, 2, 3\} \tag{1.16}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.14). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

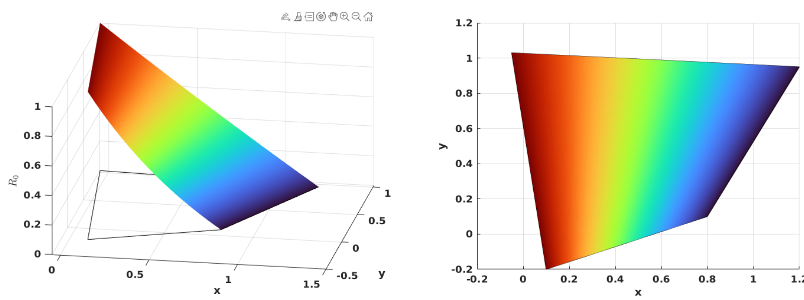


Figure 1.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.17}$$

where $\mathbf{x}_{e,i}$ is the coordinates of the middle point of the edge e_i. Here, $i \equiv i \pmod 4$. We can restrict the edge basis function to edges as

$$\varphi^{(2)}_{e,i}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j \\ 0, & i \neq j \end{cases} \tag{1.18}$$

In Figure 1.4, we present $\varphi^{(2)}_{e,0}$ on a general quadrilateral element, where $\varphi^{(2)}_{e_0}$ evalues 0 on the edges $\{e_i, \quad i \neq 0\}$, and evaluates a quadratic bubble function on the edge e_0.

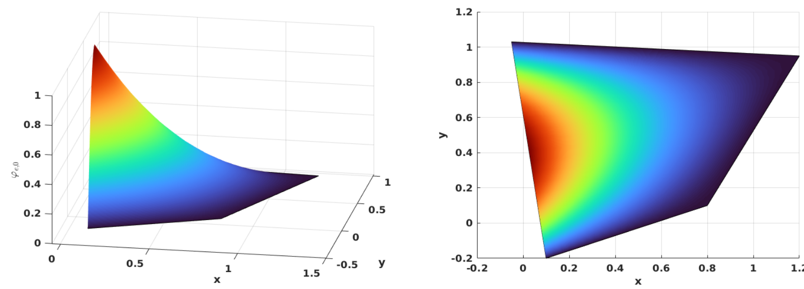


Figure 1.4: Edge nodal basis function $\varphi^2_{e,0}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^2_{v,i}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}_{v,i})}, \tag{1.19}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. In Figure 1.5, we show $\varphi^2_{v,0}$ on a general quadrilateral.

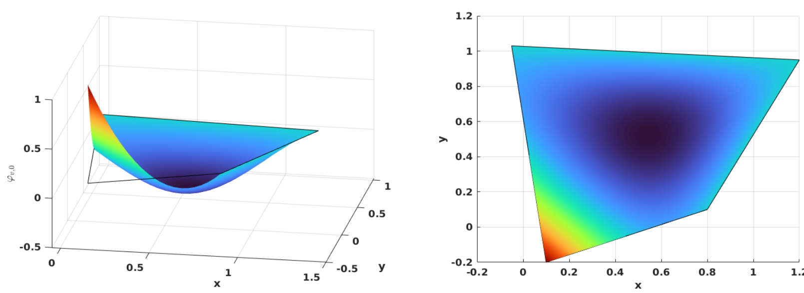


Figure 1.5: Vertex nodal basis function $\varphi^2_{v,0}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.3.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \text{span}\{R\}, \tag{1.20}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational funcion R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.17) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}), \tag{1.21}$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.22}$$

In Figure 1.6, we plot the rational function on a general quadrilateral.

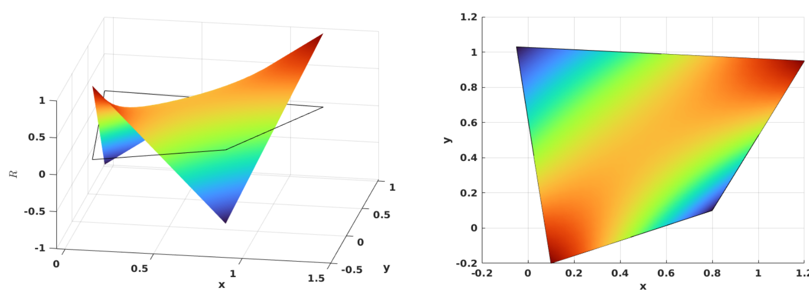


Figure 1.6: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\varphi^1_{v,i}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.23}$$

where $m \equiv i+1 \pmod 2$, and index $i \equiv i \pmod 4$. In Figure 1.7, we show the vertex nodal basis function $\varphi^{(1)}_{v,0}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

1.3.3 An Interpolation Operator

We refer to an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$ defined in Arbogast et al. (2022). Here, K_i has different meanings with different types of

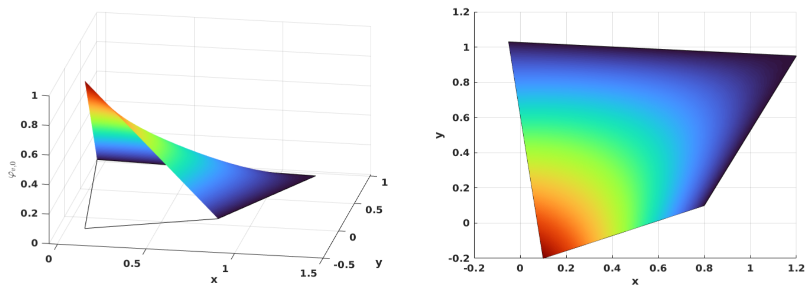


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

basis functions. For each basis functions associated with edge node, we set $K_i = e_i$. For each basis functions associated with vertex node, we set K_i being a set of edges sharing the vertex v_i. An interpolation operator $\mathcal{I}_h^r : W_p^l(E) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y}) d\mathbf{y} \in \mathcal{DS}_r, \tag{1.24}$$

where $1 \leq p \leq \infty$ and $l > 1/p$ (but $l \geq 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x}) d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \ldots, N_r\}, \tag{1.25}$$

where N_r is the number of nodal basis defined on K_i. We present the following two lemmas regarding the constructed interpolation operator for future reference.

Lemma 1.3. *(Boundedness of the interpolation operator.)*

Let $\mathcal{T}_h$ be uniformly shape regular (1.9) *with shape regularity parameter σ_*. For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator* (1.24). *For every $E \in \mathcal{T}_h$, suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1}. Moreover, assume that $R_{i,i+2}$ is uniformly differentiable functions of the vertices of E up to order m. Then for $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any nonnegative integer m,*

$$\|\mathcal{I}_h^r v\|_{W_q^m} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.26}$$

where $E^* = \cup_{F\in\mathcal{T}_h, F\cap E\neq\emptyset} F$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives defined in Definition 1.2.

Proof. See Arbogast et al. (2022). □

Combining Lemma 1.3 and Bramble-Hilbert lemma 1.2, we give local and global error estimation.

Lemma 1.4. *(Approximability of the interpolation operator.)*

Under the assumptions of 1.3, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \le p \le \infty$ and $l > 1/p$ or ($l \ge 1$ if $p = 1$),

$$\|v - \mathcal{I}_h^r v\|_{W_p^m} \le C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \le m \le \min(l, r+1). \tag{1.27}$$

Morevoer, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E\in\mathcal{T}}$, such that for all functions $v \in W_p^l(\Omega)$,

$$(\sum_{E\in\mathcal{T}_h} \|v - \mathcal{I}_h^r v\|_{W_p^m}^p)^{1/p} \le C h^{l-m} |v|_{W_p^l(\Omega)}, 0 \le m \le \min(l, r+1). \tag{1.28}$$

Proof. See Arbogast et al. (2022). □

1.4 Mixed Finite Elements

We then construct mixed finite elements from the introduced direct serendipity space. Now recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.4. A triple $(E, X(E), \varphi_j)$ defines a finite element.

- Element $E \in \mathcal{T}_h$;
- Space of shape function $X(E)$, and a set of shape function $\varphi_i \in X(E)$. The dimension of the shape function space $\dim(X(E)) = n$;

- A set of continuous and linear functionals ϕ_j

$$\phi_j : X(E) \supset \mathfrak{X}(E) \to \mathbb{R}, \tag{1.29}$$

such that unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \dots, n\} \tag{1.30}$$

In the following sections, we state our shape function space constructed using first and second order direct serendipity space and corresponding degrees of freedom. We also provide explicit construction of shape functions.

We introduce a H^1 and a $H(\text{div})$ conforming mixed element space constructed with the pre-defined direct serendipity space for Darcy and Stokes problems.

1.4.1 A Darcy Model Problem

Consider a model problem for Darcy equation as

$$\begin{aligned} \mathbf{u} + \nabla p = \mathbf{f}, &\ \text{ in } \ \Omega \\ \nabla \cdot \mathbf{u} = 0, &\ \text{ in } \ \Omega \\ \mathbf{u} = \mathbf{0}, &\ \text{ on } \ \partial\Omega \end{aligned} \tag{1.31}$$

We seek solution in the following energy space as

$$\begin{aligned} \mathbb{V}_D &= \{\mathbf{v} \in (H(\text{div};\Omega))^2 : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \ \text{ on } \ \partial\Omega\}, \\ \mathbb{U}_D &= \{\mathbf{u} \in (H(\text{div};\Omega))^2 : \mathbf{u} \cdot \boldsymbol{\nu} = \mathbf{0} \ \text{ on } \ \partial\Omega\} = \mathbb{V}_D, \\ \mathbb{W}_D &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{1.32}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_D$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), &\ \ \forall \mathbf{v} \in \mathbb{V}_D, \\ (\nabla \cdot \mathbf{u}, \omega) = 0, \qquad &\ \ \forall \omega \in \mathbb{W}_D. \end{aligned} \tag{1.33}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{D,h} \subset \mathbb{V}$ and $\mathbb{W}_{D,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned} (\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), \quad & \forall \mathbf{v}_h \in \mathbb{V}_{D,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) = 0, \quad & \forall \omega_h \in \mathbb{W}_{D,h}. \end{aligned} \tag{1.34}$$

Now we construct a pair of discretized space $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem in the next section.

1.4.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \tag{1.35}$$

$$\cup \qquad\qquad \cup \qquad\qquad \cup \tag{1.36}$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \tag{1.37}$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}. \tag{1.38}$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. The dimension of the derived space is

$$\dim(\mathbb{V}_1^0) = 2 \times 3 + 2 = 8. \tag{1.39}$$

We need two independent basis functions on each edge of a quadrilateral E, one of which has a vanishing average flux on the edge and the other one has a non-vanishing average flux on the edge.

We state the degree of freedom of element as normal flux on each edge as

$$\int_{e_i} \mathbf{u} \cdot \boldsymbol{\nu} \varphi_i d\mathbf{x}, \quad \varphi_i \in \mathbb{V}_1^0. \tag{1.40}$$

We need two independent shape functions on each edge. We present a construction of two independent basis functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.41}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.42}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we write curl of the auxiliary rational function R_i (1.14) as

$$\mathrm{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.43}$$

where the index i is understood with modulo of 4. We first present a construction of basis function $\boldsymbol{\varphi}_{l,i}$ as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \mathrm{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\mathrm{curl} R_i. \tag{1.44}$$

We then normalize it such that l_i varying from -1 to 1 as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.45}$$

where the index i is understood in the sense of modulo 4, and edge length $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

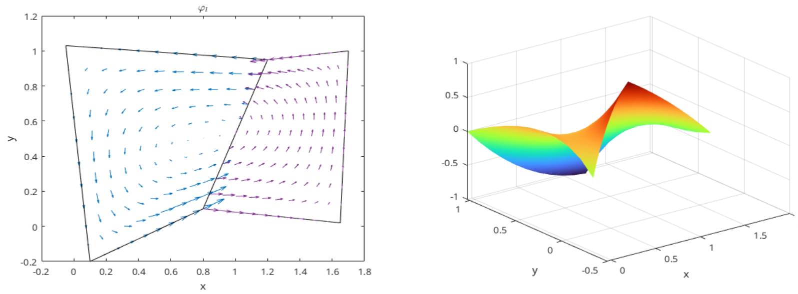


Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \big[\frac{1}{2}\varphi^2_{e,i} + \frac{1}{2}\varphi^2_{e,i+1}\big], \tag{1.46}$$

where the index i is understood in modulo of 4. Then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \mathrm{curl}\phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.47}$$

We then define the "constant" basis function defined on the edge as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.48}$$

where the index i is again understood in the sense of modulo of 4. The defined basis function $\boldsymbol{\varphi}_{c,i}$ has flux of constant value 1 on e_i and 0 on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show normal flux joining continuously as a constant 1 on the shared edge.

1.4.3 Stability and Approximation Properties for $\mathbb{V}^0_1$

The accuracy and stability of such element has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness.

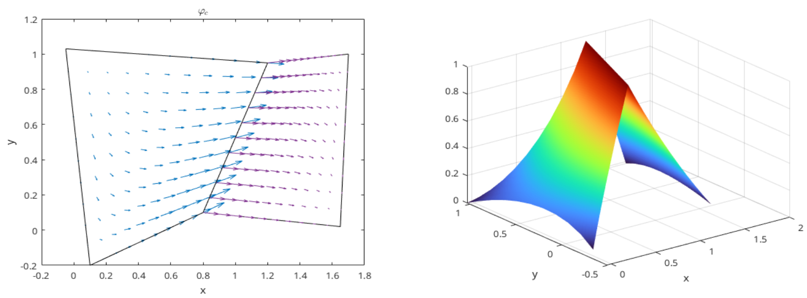


Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

Consider a projection operatior π_D preserves element average divergence and average edge normal fluxes.

$$\pi_D : H(\mathrm{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}_1^0, \tag{1.49}$$

where $\epsilon > 0$. The global projection operator π is understood as combination of local operator π_E according to degree of freedom. The operator π also satisfies the commuting diagram property as

$$\mathcal{P}_{W_s} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{1.50}$$

where $\mathcal{P}_{W_s}$ is the L^2-orthogonal projection operator onto $W_s = \nabla \cdot \mathbb{V}_1^0$, which is a constant value in our case.

Lemma 1.5. *Stability and approximation properties.*

Under the assumption of 1.3, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\|p - p_h\|_{m,\Omega} \leq C h^{l+1-m} |p|_{l+1,\Omega}, \quad l = 0, 1, 2, \ldots, r, \tag{1.51}$$

where $p_h \in \mathcal{DS}_2$. We have

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, & k &= 1, \ldots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \mathbf{u}\|_{k,\Omega} h^k, & k &= 0, \ldots, 1 \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, & k &= 0, \ldots, 1 \end{aligned} \tag{1.52}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_s, \tag{1.53}$$

holds for some $\gamma > 0$ *independent of* $\mathcal{T}_h$ *and* $h > 0$.

Proof. See Arbogast et al. (2022). □

1.4.4 A Stokes Model Problem

Consider a model problem for Stokes equation as Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.54}$$

We pick energy space for displacement and pressure as

$$\begin{aligned} \mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \ \text{ on } \partial\Omega\} = (H_0^1(\Omega))^2, \\ \mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \ \text{ on } \partial\Omega\} = \mathbb{V}_S, \\ \mathbb{W}_S &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}, \end{aligned} \tag{1.55}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \ \ \forall \mathbf{v} \in \mathbb{V}_S, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad \forall \omega \in \mathbb{W}_S. \end{aligned} \tag{1.56}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \ \ \forall \mathbf{v}_h \in \mathbb{V}_{S,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) &= 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}. \end{aligned} \tag{1.57}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ for this Stokes model problem in the next section.

1.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A similar type of element was discussed by Arbogast and Wheeler (2005).

Inspired by the work in direct serendipity space, we consider the following modified space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \mathrm{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \tag{1.58}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i. One natural pick is the previously defined function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}^{(2)}(\mathbf{x})\mathbf{n}_i, \tag{1.59}$$

which is quadratic when restricted to edge e_i according to equation (1.18). The quadratic bubble is guaranteed to be conforming since it evaluates zeros on two vertex and 1 at the middle point of the edge due to normalization. The integral of this quadratic bubble can be accurately computed by a three points gaussian quadrature rule.

Definition 1.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:

(i) for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(\mathrm{i})}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(\mathrm{i})}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF^{(ii)}_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

We give an explicit construction of basis functions using driect serendipity space. We pick nodal basis on vertex v_i in two Cartesian direction as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.60}$$

We pick edge basis on edge e_i as

$$N_e = \varphi^{(2)}_{e,i}\boldsymbol{\nu}_i \tag{1.61}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We define the corresponding discrete pressure sapce as

$$\mathbb{W}_0 = \{\omega \in L^2\Omega; \omega|_E \in P_0\} \cap L^2(\Omega)/\mathbb{R}, \tag{1.62}$$

which holds piece wise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next section.

1.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyze the stability and approximation properties for the modified, we state some useful lemma.

Lemma 1.6. *Let K be a nondengenate convex quadrilateral. Assuming we have shape regularity 1.3. For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1, and C_2 independent of the geometry of E such that the following upper bound hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(K)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{K})}, \\ |v|_{W_p^1(K)} \leq C_2(\sigma^*, p) |\tilde{v}|_{W_p^1(\tilde{K})}, \end{cases} \tag{1.63}$$

where $\tilde{K}$ represents quadrilateral before scaling illustrated in Figure 1.1, and $\tilde{v}$ is function defined on $\tilde{K}$ before scaling.

Proof. See Girault and Raviart (2011). □

With the interpolation operator $\mathcal{I}_{\mathcal{DS}_1}$ defined in 1.24, we define the projection operator $\pi_h \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{BR} \cap (H_0^1(\Omega))^2$ by:

(i) For each corner point $v \in \partial E$

$$\pi_h \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_{\mathcal{DS}_1} \mathbf{v}(\mathbf{x}_v). \tag{1.64}$$

(ii) On each edge $e \subset \partial E$,

$$\int_e (\pi_h \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} ds = 0. \tag{1.65}$$

Lemma 1.7. *For all $\mathbf{v} \in (H^k(\Omega))^2, \quad k = 1, 2$. The operator π_h defined by* (1.64) *and* (1.65) *satisfies:*

$$(p_h, \nabla \cdot (\mathbf{v} - \pi_h \mathbf{v})) = 0, \quad \forall p_h \in \mathbb{W}_0 \tag{1.66}$$

Assuming shape regularity defined in 1.3, π_h should have the error bound as

$$\|\mathbf{v} - \pi_h \mathbf{v}\|_{1,\Omega} \leq Ch^m \|\mathbf{v}\|_{m+1,\Omega}, \quad m = 0, 1. \tag{1.67}$$

Proof. From the definition of degree of freedom (1.64) and (1.65), we decompose the projection operator π_h as

$$\pi_h \mathbf{v} = \mathcal{I}_{\mathcal{DS}_1} \mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.68}$$

where the coefficient

$$\alpha_i = \Big(\int_{e_i} (\mathbf{v} - \mathcal{I}_{\mathcal{DS}_1} \mathbf{v}) \cdot \boldsymbol{\nu}_i d\sigma \Big) / \int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i d\sigma. \tag{1.69}$$

We use the approximation property of interpolation operator in direct derendipity space in Lemma 1.4,

$$\|\mathbf{v} - \mathcal{I}_{\mathcal{DS}_1} \mathbf{v}\|_{m,\Omega} \leq h_E^{2-m} |\mathbf{v}|_{2,E^*}, \quad m = 0, 1, \tag{1.70}$$

where $E^* = \mathbb{U}_{F \in \mathcal{T}_h, F \cap E \neq}$ defined in Lemma 1.3.

Lemma 1.6 indicates :

$$|\mathbf{b}_i|_{m,E} \leq C(\sigma^*) h_E^{1-m} |\tilde{\mathbf{b}}_i|_{1,\tilde{E}} \leq C(\sigma^*) h_E^{1-m}, \quad m = 0, 1. \tag{1.71}$$

We continue to estimate $|\alpha_i|$ as

$$\int_{e,i} \varphi_{e,i}^{(2)} ds = |e_i| \int_{\tilde{e}_i} \tilde{\varphi}_{e,i}^{(2)} d\tilde{s}. \tag{1.72}$$

Then

$$|\alpha_i| \leq C \int_{\tilde{e},i} |\tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_{\mathcal{D}\mathcal{S}_1} \tilde{\mathbf{v}}| ds \tag{1.73}$$

Trace theorem, poincare inequality ...

The final estimate is then

$$|\alpha_i||\mathbf{b}_i|_{m,E} \leq C h^{k-m} |\mathbf{v}|_{k,E^*}. \tag{1.74}$$

$\square$

Our space $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ form a $\int$-sup stable pair.

Theorem 1.8. *For a convex domain Ω, there exists a constant $\gamma > 0$ such that*

$$\inf_{\omega \in \mathbb{W}_0} \sup_{\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}, \omega)}{\|\mathbf{v}\|_1 \|\omega\|_0} \geq \gamma > 0. \tag{1.75}$$

Proof.

$$\sup_{\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}, \omega)}{\|\mathbf{v}\|_1 \|\omega\|_0} \geq \frac{(\nabla \cdot \pi_h \varphi, \omega)}{\|\pi_h \varphi\|_1 \|\omega\|_0} \geq \frac{\|\omega\|_0}{C \|\varphi_1\|} \geq \gamma > 0, \tag{1.76}$$

$\square$

where π is a bounded projection operator.

1.4.7 Test Modified Bernardi-Raugel Element

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta\mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla\cdot\mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.77}$$

We define a simple choice of energy space

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \quad \text{on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)/\mathbb{R}\}, \end{aligned} \tag{1.78}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\Omega))^d$ and understand as a continuous extension $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ from the boundary into the domain. We write the problem in weak form as:

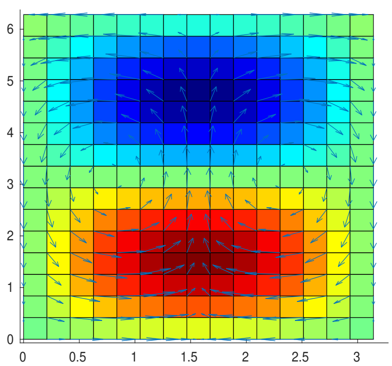
Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla\cdot\mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla\cdot\mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}. \end{aligned} \tag{1.79}$$

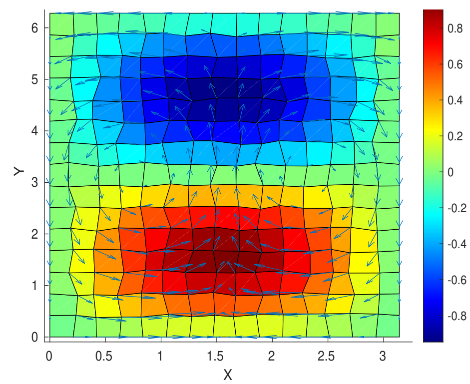
We consider the true solution of test problem (1.79) defined on a square region $\Omega = [0,\pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{1.80}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing a condition that $\int_\Omega p d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular 1.10a and logically rectangular quadrilateral 1.10b meshes. We calculate the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).



(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. Second order convergence rate is achieved by velocity and its corresponding derivatives. A first order convergence rate is achieved by pressure. Both meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\mathrm{res}_p)$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\mathrm{res}_{du})$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 1.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\mathrm{res}_p)$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\mathrm{res}_{du})$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 1.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

1.5 Coupled Degenerate Darcy-Stokes System

Consider a degenerate scaled darcy-stokes coupled boundary-value problem:

Find velocity-pressure potential pair $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.81}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.82}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.83}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.84}$$

and

$$\begin{aligned} \check{\mathbf{v}}_r &= \mathbf{g}_r \quad \text{on } \Gamma_{u,D}, \\ \check{\mathbf{v}}_s &= \mathbf{g}_s \quad \text{on } \Gamma_{u,S}, \\ p &= p_0 \quad \text{on } \Gamma_{i,D}, \\ \boldsymbol{\tau}\cdot\boldsymbol{\nu} - p &= \boldsymbol{t}_0 \quad \text{on } \Gamma_{i,S}, \end{aligned} \tag{1.85}$$

where $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the part of boundary we assign essential boundary conditions for Darcy and Stokes problem respectively, and $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the part of boundary we assign natural boundary conditions respectively. We have $\Gamma_{u,D}\cup\Gamma_{i,D} = \partial\Omega$ and $\Gamma_{u,S}\cup\Gamma_{i,S} = \partial\Omega$.

We follow the procedure by Arbogast et al. (2017) to discretize this boundary value problem. To begin weith, a new Hilbert space is defined for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in, so

$$\mathbb{V}_d^{\phi} = H_{\phi}(\mathrm{div};\Omega) = \Big\{\mathbf{v}\in(L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Phi}\mathbf{v})\in L^2(\Omega)\Big\}, \tag{1.86}$$

which equipped with the inner product as

$$(\mathbf{u}_\phi, \mathbf{v}_\phi)_{\mathbb{V}_r^\phi} = (\mathbf{u}_\phi, \mathbf{v}_\phi) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}_\phi), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}_\phi)). \tag{1.87}$$

The normal trace on $\partial\Omega$ is well-defined as well as

$$\big\|\phi^{1/2+\Theta}\mathbf{v}_\phi\cdot\nu\big\|_{-1/2,\partial\Omega} < C_\Omega\Big\{\,\|\mathbf{v}_\phi\| + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}_\phi)\big\|\,\Big\}. \tag{1.88}$$

1.5.1 The weak formulation

We define the following function spaces

$$
\begin{aligned}
\mathbb{V}_D^{\phi} &= \Big\{ \mathbf{v} \in H_{\phi}(\mathrm{div};\Omega) : \phi^{1/2+\Theta} \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D} \Big\}, \\
\mathbb{W}_D &= L^2(\Omega), \\
\mathbb{V}_S &= \Big\{ \mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S} \Big\}, \\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned}
\tag{1.89}
$$

where each space is equipped with their natural norm. To empose essential boundary conditoins, we assume that a function $\mathbf{g}_s = \check{\mathbf{v}}_{s,0}$ on $\Gamma_{u,S}$ and $\mathbf{g}_s = \mathbf{0}$ on $\partial\Omega/\Gamma_{u,S}$. Then $\mathbf{g}_s \in (H^{1/2}(\partial\Omega))^2$ and extended it continuously from the boundary to the domain, so that the extension $\mathbf{g}_s \in (H^1(\Gamma_{u,S}))^2$ and $\|\mathbf{g}_{s,0}\|_1 \leq C \|\mathbf{v}_{s,0}\|_{1/2,\Gamma_{u,S}}$. Similarly, we can define a bounded extension of $\check{\mathbf{v}}_r \in \mathbb{V}_D^{\phi}$ such that

Scaled nondimensionalized weak form. Find $\check{\mathbf{v}}_r \in \mathbb{V}_D^{\phi} + \tilde{\mathbf{g}}_r$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{V}_S + \tilde{\mathbf{g}}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$
(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall \boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{1.90}
$$

$$
(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall \omega \in \mathbb{W}_S, \tag{1.91}
$$

$$
(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)) = 0, \qquad \forall \boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{1.92}
$$

$$
(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_f\Big) = 0 \qquad \forall \omega_f \in \mathbb{W}_D. \tag{1.93}
$$

The Existence and uniquense of the solution has been proved by Arbogast et al. (2017). We will not repeat the proof here.

1.5.2 MFEM discretization

We discretize the formed degenerate system with previously defined $H(\text{div})$ and H^1 conforming space. We write the function spaces out as

$$
\begin{aligned}
\mathbb{V}_1^0 &= (\mathbb{P}_1)^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}} \\
\mathbb{W} &= \mathbb{P}_0 \\
\mathbb{V}_1^{\mathcal{BR}} &= \mathcal{DS}_1 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \\
\mathbb{W} &= \mathbb{P}_0.
\end{aligned}
\tag{1.94}
$$

We can impose essential boundary conditions

Scaled mixed finite element method. Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_1^0 + \hat{\mathbf{g}}_r$, $\check{q})_{r,h} \in \mathbb{W}_D$, $\mathbf{v}_{s,h} \in \mathbf{V}_S + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_S$ such that

$$
(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s) - (\check{q}_h, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall \boldsymbol{\psi}_s \in \mathbb{V}_1^{\mathcal{BR}}, \tag{1.95}
$$

$$
(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega\Big) = 0 \qquad \forall \omega \in \mathbb{W}_0^{\mathcal{BR}}, \tag{1.96}
$$

$$
(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r) - (\check{q}_{l,h}, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_r \in \mathbb{V}_1^0, \tag{1.97}
$$

$$
(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_f\Big) = 0 \qquad \forall \omega_f \in \mathbb{W}_0. \tag{1.98}
$$

1.5.3 Local Mass Conservation

Before we continue the discussion, we need to make sure the formulation is well-defined by not dividing by zero. The term $\phi^{-1/2}$ in the symmetric equations (1.97) and (1.98) causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we first define a element-averaged $\overline{\phi}_E$ as

$$
\overline{\phi}_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})d\mathbf{x}, \tag{1.99}
$$

where $|E|$ is the area of element E. Let

$$
\phi_E = \begin{cases} \overline{\phi}_E & \text{if } \overline{\phi}_E \neq 0, \\ 0 & \text{if } \overline{\phi}_E = 0. \end{cases} \tag{1.100}
$$

We now replace the $\phi^{-1/2}$ with $\phi_E^{-1/2}$. Then we expand the integral with divergence theorem as

$$\begin{aligned}\int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_l = &\int_{\partial E}\phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu}\omega_f ds \\ &-\int_E \phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\nabla\omega_l d\mathbf{x}.\end{aligned} \tag{1.101}$$

The second term vanishes when ω_l is constant on element E. We define an auxiliary auxiliary space ω_E begin 1 on E and 0 elsewhere. We sum equations (1.96) and (1.98) together assuming that $\omega = \omega_E \in \mathbb{W}_S$ and $\omega_l = \phi_E^{-1/2}\omega_E \in \mathbb{W}_D$. We can achieve local mass conservation of the phase averaged mixture if we replace the term $\phi^{1/2}$ in equation (1.96) with $\phi_E^{1/2}$ as

$$\int_E \nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r + \check{\mathbf{v}}_s)d\mathbf{x} = 0. \tag{1.102}$$

We now propose our local mass conserved version as

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}),\nabla\boldsymbol{\psi}_s) - (\check{q}_h,\nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n},\boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_1^{\mathcal{BR}}, \tag{1.103}$$

$$(\nabla\cdot\check{\mathbf{v}}_{s,h},\omega) - \left(\frac{\phi_E^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h),\omega\right) = 0 \qquad \forall\omega \in \mathbb{W}_0^{\mathcal{BR}}, \tag{1.104}$$

$$(\check{\mathbf{v}}_{r,h},\boldsymbol{\psi}_r) - (\check{q}_{l,h},\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_1^0, \tag{1.105}$$

$$(\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}),\omega_f) + \left(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h),\omega_f\right) = 0 \qquad \forall\omega_f \in \mathbb{W}_0. \tag{1.106}$$

1.5.4 Convergence Rate

We derive a bound for the error by taking the difference of

1.6 Saddle Point System

Saddle point system is widely spread in technical and science scenarios, such as constrained optimization problem, image reconstruction and optimal control. Saddle point system has been widely and systematically studied, e.g., by Benzi et al. (2005). In our case, saddle point system is formed along with the Mixed Finite Element discretization. In this section, we give a brief overview of properties of a saddle point system and some common ways to solve such system. An abstract saddle point system can be represented as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{g} \end{bmatrix}. \tag{1.107}$$

Assuming the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

1.6.1 Reduced Linear System

Now we discuss the saddle point system (SPS) induced by MFEM method in details including the boundary conditions. To begin with, we regroup dofs according to the boundary condition.

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{1.108}$$

where $\mathbf{A}_K$ consists of interaction within interior dofs and natural boundary condition dofs, and $\mathbf{A}_M$ consists of iteraction between interior dofs, natural boundary dofs and essential boundary dofs.

We perform the same treatment for matrix $\mathbf{B}$ and right hand side vector $\mathbf{f}$, by regrouping dofs according to its iteration with boundary conditions.

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix}, \text{and} \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix} \tag{1.109}$$

We rewrite the linear system with the regrouped dofs and form a reduced system as

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M\mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T\mathbf{g} \end{bmatrix}, \tag{1.110}$$

where $\mathbf{g}$ denotes the values prescribed for essential boundary, and $\mathbf{g}_N$ is the vector generated by the natural boundary condition part. The reformed reduced system is still a saddle point syste.

Now we take a closer look at the formed linear system. The no melting (zero porosity $\phi = 0$) region adds difficulties to solving the formed saddle point system. On one hand, the existence of zero porosity region guarantees that $\mathbf{B}$ matrix formed in Darcy system does not has full column rank. Therefore the schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T),$$

is usually not invertable. On the other hand, in addition to the constant null space induced by gradient of pressure, the null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of zero porosity region).

Taking into consideration that the solver will be implemented in parallel, a iterative solver is more suitable than direct solver.

1.6.2 Uzawa Iteration

Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), which gained popularity in computational fluid dynamics. It has later been applied to the case of discretizing Stokes problem with Mixed Finite Element. Some inexact inner solver to accelerate uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We start by stating an abstract inexact ezawa iteration algorithm in Algorithm 1. Some choose diagonal matrix $\alpha\mathbf{I}$ for precondictioner $\mathbf{Q}_A$. However, such simple matrix wouldn't generate a convergence result when applied to our coupled problem. In the next section, we introduce a modified algorithm that would solve our system.

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$

2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

3: Obtain $\mathbf{r} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$

4: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{r}$

1.6.3 Modified Uzawa Iteration

The linear system formed from the equations (1.103), (1.104), (1.105) and (1.106) is

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.111}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.112}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.113}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.114}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system.

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$

2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$

4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$

5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

1.6.4 Test Run

We test our MFEM setup with two problems. First, we test with a manufactured solution. Second, we test a more complex yet a more realistic problem.

Problem setup 1:

We solve a manufactured system of equations on the rectangular domain $(x, y) \in [-1, 1]^2$. We assume a constant porosity acorss the domain as ϕ. We compute the right hand side with prescribed functions as

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{1.115}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{1.116}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{1.117}$$

Substitute the compuated results into PDE

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla\check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\
\phi_f^{1/2}\nabla\cdot(\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\
-2(1-\phi_f)\nabla\cdot\sigma(\check{\mathbf{v}}_s) = -2(1-\phi_f)\nabla\cdot(\mathcal{D}\check{\mathbf{v}}_s &- \frac{1}{3}(\nabla\cdot\check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix}
\end{aligned}
\tag{1.118}
$$

Problem setup 2:

We solve the full system of equations on the rectangular domain $-160\text{km} < x < 160\text{km}$, $0, x, 160\text{km}$. The boundary conditions are define by imposing

$$
\mathbf{v}_s = \tag{1.119}
$$

1.7 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitriary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general conservation law retarding a scalar variable u as

$$
\begin{cases}
\dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla\cdot\mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x}\in\Omega, \\
u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x}\in\Omega,
\end{cases}
\tag{1.120}
$$

where we assume the domain $\Omega\subset\mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$
\frac{\partial\overline{u}}{\partial t} + \frac{1}{|E|}\int_{\partial E}\mathbf{F}(\mathbf{x},t)\cdot\boldsymbol{\nu}d\mathbf{x} = 0, \tag{1.121}
$$

where we use the cell-averaged value $\overline{u} = \frac{1}{|E|}\int_E u(\mathbf{x},t)d\mathbf{x}$, and $F(\mathbf{x},t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+,u^-) = \frac{1}{2}[(\mathbf{f}(u^+)+\mathbf{f}(u^-))\cdot\boldsymbol{\nu} - \alpha_{LF}(u^+ - u^-)], \tag{1.122}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Fredrich stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global LF Stabilizer: $\alpha_{LF} = \max_u|\partial(\mathbf{f}\cdot\boldsymbol{\nu})/\partial u|$;
- The local LF Stabilizer: $\alpha_{LF} = \max(|\partial(\mathbf{f}^+\cdot\boldsymbol{\nu})/\partial u^+|, |\partial(\mathbf{f}^-\cdot\boldsymbol{\nu})/\partial u^-|)$.

The numerical flux $\mathbf{F}$ (1.122) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+,\mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+)+\mathbf{f}(\mathcal{R}^-)\cdot\boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)], \tag{1.123}$$

where $\mathcal{R}$ is a higher order reconstruction value. We evaluate the integrals over each edge $e_i \in \partial E$ with a quadrature rule. We represent length of edge e_i and area of cell E with $|e_i|$ and $|E|$ respectively. Assuming a quadrature rule with G points on each edge, we have

$$\overline{u}_t + \sum_{i=0}^{4}\frac{|e_i|}{|E|}\sum_g \omega_{i,g}\mathbf{F}(\mathcal{R}^+_{i,g},\mathcal{R}^-_{i,g}) = 0, \tag{1.124}$$

where $\omega_{i,g}$ represents gth quadrature points on edge e_i.

1.8 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of discontinuous values with cell averaged values in Arbogast et al. (2024). The ML-WENO reconstruction will be primarily used in reconstruction of values on both sided of the edge for numerical flux computation appears in FVM discretization. It can also be applied to general reconstruction of discontinuous variables, e.g., discontinuos darcy pressure appears on the zero-nonzero porosity interface.

1.8.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \ldots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{1.125}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E \in \mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k / k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{1.126}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{1.127}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (1.127) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{1.128}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|}\int_{E_{k'}} \phi^k_{s,t}(\mathbf{x})d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|}\iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi,\eta)d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \tag{1.129}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi,\eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi,\eta). \tag{1.130}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980a) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space. For simplicity, we assume $r = s = t$.

Lemma 1.9. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{1.131}$$

Proof.

$$\|u - \pi_S u\| = \tag{1.132}$$

□

In a general 2D situtation when $s \neq t$, the exececssive order will be "wasted", since the order or accuracy will be $\min(s,t)$. However, in some cases, where we aquire the prior knowledge of the solution being qusi-1D, e.g., the transport velocity is fixed in one direction. We can have uneven stencils that has $\max(s,t)$ order of accuracy but in the desired direction. In other cases when derivative is involved, we also need an stretched stencil to compensate for the order loss when approximates derivative.

1.8.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1\leq|\alpha|\leq m} \int\int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x,y))^2 dxdy, \tag{1.133}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1}\partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_P &= \sum_{1\leq|\alpha|\leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\big)^2 d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'.j'} \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{1.134}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

We proposed an alternative way of defining smoothness indicator in Huang et al. (2023).

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{1.135}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the compuational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases} \tag{1.136}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

1.8.3 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (1.130)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{1.137}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil

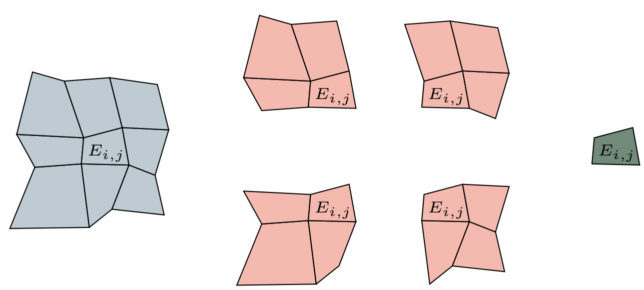


Figure 1.11: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{1.138}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e-4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{1.139}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{1.140}$$

1.8.4 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \dots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \ \ u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \ \ u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{1.141}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s, t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s, t) that has a jump as $\mathcal{L}_{s,t} \cap$ Jump. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{1.142}$$

In Figure 1.12, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0, 1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump

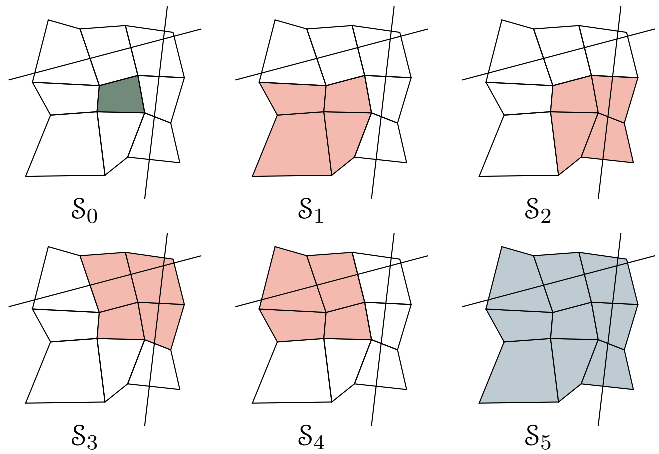


Figure 1.12: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

We state the estimation of reconstruction error show in Arbogast et al. (2024). We start by estimating the error of our nonlinear scaling (1.138) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{1.143}$$

We estimate the reconstruction error for two situtaions.

- $l \in \text{Smooth}$, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in \text{Jump}$, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from Olver (1997). Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{1.144}$$

For the smooth stencil $\mathcal{S}_l \in$ Smooth

$$\begin{aligned}\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\ &= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1+\mathcal{O}(h))\Big)^m \\ &= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)).\end{aligned} \tag{1.145}$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{1.146}$$

Then we estimate the error for the normalized nonlinear weighting (1.140). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{1.147}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1+\mathcal{O}(h)). \tag{1.148}$$

Assume $h < 1$, we keep the largest $ar + \eta(r)$ number as our estimator

$$\begin{aligned}\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = &\sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\ &+ \Theta(h^{-2(ar+\eta(r))}) \\ = &\sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\ &+ \mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}).\end{aligned} \tag{1.149}$$

later, we denote the set Smooth $\cap \mathcal{L}_r$ as Smooth_r for simplicity. The overall accuracy estimation of the nonlinear weighting (1.140) is presented as

- $l \in$ Smooth

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l' \in \text{Smooth}_r} \omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{1.150}$$

- $l \in$ Jump

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{1.151}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}
|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l (u(\mathbf{x} - Q_l(\mathbf{x})))| \\
&\leq \sum_{l \in \text{Smooth}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| + \sum_{l \in \text{Jump}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| \\
&\leq \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\
&\qquad + \sum_{l \in \text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\
&= \mathcal{O}(h^{r_{\max}})
\end{aligned} \tag{1.152}$$

Combining everything above and (1.137), we arrive at the formal error estimate of the reconstruction.

Lemma 1.10. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \leq Ch^{r_{\max}}, \tag{1.153}$$

where $r_{\max}$ is defined by (1.142).

1.8.5 Handling Boundary Conditions

In our simulation, we couple FEM and FVM solver on a single mesh. Therefore, accurate and compatible boundary condition should be assigned accordingly. However, it is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points. The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

In order to assign proper boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no fixed boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

The inflow boundary can be understood in various ways. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B as an equivelent to Dirichlet boundary in the context of Finite Element. In the special case when a zero flux is prescribed, we would call it noflow boundary. In the inflow case, there may be a discontinuous inflow shock wave. We require a swift response inside the domain to align with the specified Dirichlet value. We therefore add an extra constant stencil polynomial to the inflow boundary to accommodate the potential discontinuity.

There are some other special cases that require special treatment on the boundary, e.g., when symmetry should be enforced.

1.8.6 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x, y) = 1$, and the computational domain

$\Omega = [0, 3] \times [0, 1]$. The true solution is

$$u(x, y, t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x - 0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{1.154}$$

We forward martching to the time $= 2.0$, when trailing rarefaction reaches the leading shock. We use SSP2RK time stepping scheme.

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